IGCSE 0607 Practice Questions — E1.14 — Combined Questions & Markscheme

Time

Total Questions: 21 — **Total Marks:** 115 Marks

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  → **Paper 2 (Non-Calculator)** *(7 Qs)* · **Paper 4 (Calculator)** *(14 Qs)*

1 A train journey from Portsmouth to London is shown in the timetable.

Station	Time
Portsmouth	08 45
London	11 20

(a) Work out the journey time in hours and minutes.

hours ..... , minutes ..... **[1]**

(b) The return journey takes exactly the same amount of time.
The train leaves London at 3.35 p.m.

Write down the time it arrives in Portsmouth using the 24-hour clock.

..... **[2]**

2 A train timetable shows that a train leaves Paris at 09 55 and arrives in Lyon at 12 10.

(a) Work out the time taken for this journey in hours and minutes.

hours ..... , minutes ..... **[1]**

(b) Another train leaves Paris at 16 45.
This journey takes 2 hours and 35 minutes.
Work out the time this train arrives in Lyon.

..... **[2]**

3 A flight departs from London to Dubai at 21 : 45 local time.

The flight time is 6 hours and 50 minutes.
The local time in Dubai is 4 hours ahead of the local time in London.
Work out the local time in Dubai when the flight arrives.

..... **[3]**

4 A flight leaves Dubai at 14 30 local time on Friday and arrives in London at 19 15 local time on the same day.
The local time in Dubai is 4 hours ahead of the local time in London.

(a) Work out the duration of the flight in hours and minutes.

hours, minutes **[2]**

(b) A second flight leaves Dubai at 22 45 local time on Friday.
The duration of this flight is 7 hours and 20 minutes.
Work out the local time and the day it arrives in London.

Time = ... , Day = ... **[2]**

5 **(a)** A train departs from Paris at 22:50.
The journey to Rome takes 14 hours and 35 minutes.
Work out the arrival time in Rome using the 24-hour clock.

..... **[2]**

(b) Convert 3.4 hours into hours and minutes.

h ..... , min ..... **[2]**

6 The timetable shows part of a train journey.

Station	Arrive	Depart
Alphaville	–	09 45
Betatown	11 20	11 35
Gammacity	14 15	–

(a) Calculate how long the journey from Alphaville to Betatown takes.

hours ..... , minutes ..... **[2]**

(b) A later train leaves Alphaville at 14 55.
The journey time to Gammacity is exactly 4 hours and 10 minutes.
At what time does this train arrive at Gammacity?

..... **[2]**

7 **(a)** A movie starts at 19 35 and lasts for 115 minutes.
Work out the time the movie ends in the 24-hour clock.

..... **[2]**

(b) A plane leaves New York at 21 40 local time.
The flight to Paris takes 7 hours and 15 minutes.
The local time in Paris is 6 hours ahead of New York.
Find the local time in Paris when the plane arrives.

..... **[3]**

8 A flight departs from Dubai at 14 30 local time and arrives in Sydney.
The flight time is 14 hours and 15 minutes.
The local time in Sydney is 7 hours ahead of the local time in Dubai.
Work out the local time in Sydney when the flight arrives.

..... **[3]**

9 Light travels at 3×10^5 km/s.

(a) Calculate the distance light travels in one year (365 days).
Give your answer in standard form.

..... km **[2]**

(b) The distance from Earth to a star is 4.5×10^{13} km.
Calculate the time taken for light to travel from the star to Earth.
Give your answer in years, correct to 3 significant figures.

..... years **[2]**

10 A train travels from Station A to Station B. The total distance is 420 km.
The train departs Station A at 08 : 15 and arrives at Station B at 11 : 03.

(a) Calculate the time taken for the journey, in hours and minutes.

hours, minutes **[2]**

(b) Calculate the average speed of the train in km/h.

..... km/h **[3]**

11 Dubai is in the time zone UTC+4 and London is in the time zone UTC+0.

(a) An airplane departs Dubai at 02 15 local time on Tuesday and flies to London.
The flight takes 7 hours and 45 minutes.

Find the local time and day in London when the flight arrives.

Time ..... , Day ..... **[3]**

(b) An astronaut spends exactly 1.5 years in space.
Given that 1 year = 365 days, calculate the total number of minutes the astronaut spends in space.

Give your answer in standard form.

..... minutes **[3]**

12 **(a)** A flight from Tokyo to London takes 14 hours and 15 minutes.
Tokyo is 9 hours ahead of London.
The flight arrives in London at 16:30 local time on Tuesday.
Calculate the departure time from Tokyo in local time.

..... **[3]**

(b) A machine completes one cycle every 1 minute and 45 seconds.
Calculate the maximum number of full cycles it can complete in 8 hours.

..... **[3]**

13 A flight departs from London and travels to Dubai.
The local time in Dubai is 4 hours ahead of the local time in London.

(a) The flight leaves London at 13 45 local time.
It arrives in Dubai at 00 15 local time the next day.
Calculate the total duration of the flight.

hours ..... , minutes ..... **[3]**

(b) The total distance of the flight is 5486 km.
Using your answer to part (a), calculate the average speed of the airplane in km/h.

..... km/h **[3]**

14 **(a)** A marathon runner completes a 42.2 km race at an average speed of 12.5 km/h.

Work out the time taken to complete the race.
Give your answer in hours, minutes and seconds, correct to the nearest second.

hours = ... , minutes = ... , seconds = ... **[4]**

(b) Another runner finishes the race at 13 04.
She started the race at 08 35.
Work out how long she took to complete the race, in hours and minutes.

hours = ... , minutes = ... **[2]**

15 A communications satellite orbits the Earth.
It completes one full orbit every 94 minutes and 30 seconds.

(a) Work out the number of full orbits the satellite completes in a 30-day month.

..... **[3]**

(b) The satellite passes directly over an observatory at 05 50 on 1st March.
Find the time and date when it completes its 100th orbit after this point.

Time ..... , Date ..... **[4]**

16 A factory machine produces items continuously, 24 hours a day.

(a) The machine produces items at a constant rate of 80 items per minute.
Calculate the time it takes to produce exactly 1 000 000 items.
Give your answer in days, hours and minutes.

days ..... , hours ..... , minutes ..... **[3]**

(b) A company in Tokyo orders a batch of 1 000 000 items.
The factory is located in New York.
The local time in Tokyo is 14 hours ahead of the local time in New York.
The machine starts producing this batch at 08 00 local New York time on Monday.
Calculate the local Tokyo time and day when the batch is finished.

Day ..... , Time ..... **[4]**

17 A ship travels a distance of 920 km at an average speed of 24 km/h.

(a) Work out the time taken for the journey.
Give your answer in hours and minutes.

hours ..... , minutes ..... **[3]**

(b) The ship left the port on Wednesday at 17 50.
Determine the day and the time it arrives at its destination.

Day ..... , Time ..... **[4]**

18 An aircraft flies from City A to City B, a distance of 6600 km.
The local time at City B is 5 hours behind the local time at City A.
The aircraft leaves City A at 21 40 local time on Tuesday.
The average speed of the aircraft is 800 km/h.

(a) Calculate the time of the flight in hours and minutes.

hours, minutes **[2]**

(b) Find the local time and the day when the aircraft arrives at City B.

Time = ... , Day = ... **[2]**

(c) The aircraft stays at City B for 14 hours and 20 minutes before flying back to City A.
The return flight takes 7 hours and 45 minutes.
Find the local time and the day when the aircraft arrives back at City A.

Time = ... , Day = ... **[3]**

19 A cyclist travels 35 km at an average speed of x km/h.
She then travels a further 42 km at an average speed of $(x - 5)$ km/h.

(a) Write down an expression, in terms of x, for the total time taken in hours.

..... **[1]**

(b) The total time for the journey is 4 hours and 33 minutes.
Show that $13x^2 - 285x + 500 = 0$.

[4]

(c) Solve the equation to find the value of x.

$x =$ ..... **[3]**

20 Consider the following applications of time, speed, and dates.

(a) A satellite completes one orbit of the Earth in exactly 92 minutes.
Calculate the number of full orbits it completes in exactly one non-leap year (365 days).

..... orbits **[4]**

(b) A plane flies from City P to City Q. The distance is 11 440 km and the average speed is 880 km/h.

(i) Calculate the flight time.

..... hours **[1]**

(ii) City P is in the time zone UTC−4.
City Q is in the time zone UTC+9.
The plane departs from City P on Monday at 22 : 45 local time.
Find the local day and time of arrival in City Q.

Day = ... , Time = ... **[3]**

21 **(a)** A satellite takes 92 minutes and 45 seconds to complete one orbit of the Earth.
Work out the total time taken to complete 15 orbits.
Give your answer in hours, minutes and seconds.

h ..... , min ..... , s ..... **[2]**

(b) A flight departs City A at 09:45 local time.
The flight time is 8 hours and 20 minutes.
It arrives in City B at 02:05 local time the next day.
City B is h hours ahead of City A.

Find the value of h.

$h =$ ..... **[3]**

(c) Train P leaves Station X at 08:00 and travels to Station Y at a constant speed of 90 km/h.
Train Q leaves Station Y at 08:30 and travels to Station X at a constant speed of 120 km/h.
The distance between X and Y is 360 km.
Work out the exact time when the two trains pass each other.

..... **[4]**

1 **(a)** 2 hours 35 minutes **[B1]**

(b) M1 for 15 35 or 17 70 seen

18 10 **[A1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** 2 hours 15 minutes **[B1]**

(b) Adds 2 hours 35 minutes to 16 45 (e.g., 18 80) **[M1]**

19 20 **[A1]**

[Back to Question] *[Total Marks: 3]*

3 Arrival time in London local time = 21 : 45 + 6 hours 50 minutes **[M1]**

04 : 35 (next day) **[M1]**

Dubai local time = 04 : 35 + 4 hours = 08 : 35 **[A1]**

[Back to Question] *[Total Marks: 3]*

4 **(a)** London departure time = 10 30 or Dubai arrival time = 23 15 **[M1]**

8 hours 45 minutes **[A1]**

(b) London departure time = 18 45 or Dubai arrival time = 06 05 (Saturday) **[M1]**

02 05 Saturday **[A1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** 22 50 + 14 h 35 min **[M1]**

13 25 **[A1]**

(b) 0.4×60 **[M1]**

3 h 24 min **[A1]**

[Back to Question] *[Total Marks: 4]*

6 **(a)** Time from 09 45 to 10 00 = 15 minutes, and 10 00 to 11 20 = 1 hour 20 minutes **[M1]**

1 hour 35 minutes **[A1]**

(b) 14 55 + 4 hours = 18 55 **[M1]**

19 05 **[A1]**

[Back to Question] *[Total Marks: 4]*

7 **(a)** 115 minutes = 1 hour 55 minutes **[M1]**

21 30 (or 21 : 30) **[A1]**

(b) Arrival time in New York = 21 40 + 7 h 15 m = 04 55 (next day) **[M1]**

Arrival time in Paris = 04 55 + 6 hours **[M1]**

10 55 (or 10 : 55) **[A1]**

[Back to Question] *[Total Marks: 5]*

8 Flight arrival time in Dubai local time: 14 30 + 14 h 15 min = 28 45 (or 04 45 next day) **[M1]**

Adds time zone difference: 28 45 + 7 hours (or 04 45 + 7 hours) **[M1]**

11 45 **[A1]**

[Back to Question] *[Total Marks: 3]*

9 (a) $3 \times 10^5 \times 60 \times 60 \times 24 \times 365$ **[M1]**

9.46×10^{12} or 9.4608×10^{12} **[A1]**

(b) $\dfrac{4.5 \times 10^{13}}{9.4608 \times 10^{12}}$ **[M1]**

4.76 **[A1]**

[Back to Question] *[Total Marks: 4]*

10 (a) 11 : 03 − 08 : 15 **[M1]**

2 hours 48 minutes **[A1]**

(b) 48 minutes $= \dfrac{48}{60}$ hours = 0.8 hours **[M1]**

Average speed $= \dfrac{420}{2.8}$ **[M1]**

150 **[A1]**

[Back to Question] *[Total Marks: 5]*

11 (a) M1 for arrival time in Dubai (10 00 Tuesday) OR departure time in London (22 15 Monday) seen

M1 for a valid time zone adjustment and flight duration addition

06 00, Tuesday **[A1]**

(b) M1 for $1.5 \times 365 \times 24 \times 60$

A1 for 788400

7.884×10^5 **[B1]**

[Back to Question] *[Total Marks: 6]*

12 **(a)** London departure time = 16 30 − 14 h 15 min or 02 15 **[M1]**

Tokyo departure time = 02 15 + 9 h **[M1]**

11 15 **[A1]**

(b) 1 min 45 s = 105 seconds or 8 hours = 480 minutes **[B1]**

$(8 \times 3600) \div 105$ or $480 \div 1.75$ **[M1]**

274 **[A1]**

[Back to Question] *[Total Marks: 6]*

13 **(a)** London departure time in Dubai local time = 13 45 + 4 hours = 17 45 **[M1]**

Time from 17 45 to 00 15 = (24 00 − 17 45) + 15 minutes = 6 hours 15 minutes +15 minutes **[M1]**

6 hours 30 minutes **[A1]**

(b) Flight time converted to decimal hours = 6.5 **[B1]**

Speed = $\frac{5486}{6.5}$ **[M1]**

844 **[A1]**

[Back to Question] *[Total Marks: 6]*

14 **(a)** Time = $\frac{42.2}{12.5}$ or 3.376 hours **[M1]**

$0.376 \times 60 = 22.56$ minutes **[M1]**

$0.56 \times 60 = 33.6$ seconds **[M1]**

3 hours, 22 minutes, 34 seconds **[A1]**

(b) Correct difference calculation, e.g. 12 hours 64 minutes −08 hours 35 minutes **[M1]**

4 hours 29 minutes **[A1]**

[Back to Question] *[Total Marks: 6]*

15 **(a)** M1 for $30 \times 24 \times 60$ or 43200 minutes seen

M1 for $\frac{43200}{94.5}$ or 457.142...

457 **[A1]**

(b) M1 for 100×94.5 or 9450 minutes

M1 for converting minutes into hours and days (e.g., 157.5 hours OR 6 days 13 hours 30 minutes)

M1 for adding their 6 days and 13 hours 30 minutes to 05 50 on 1st March

19 20, 7th March **[A1]**

[Back to Question] *[Total Marks: 7]*

16 **(a)** Total minutes $= \frac{1\,000\,000}{80} = 12\,500$ **[M1]**

Total hours $= 12\,500 \div 60 = 208$ with remainder 20 (so 20 minutes) **[M1]**

Total days $= 208 \div 24 = 8$ with remainder 16 (so 16 hours) **[M1]**

8 days, 16 hours, 20 minutes **[A1]**

(b) Production start in Tokyo time = 08 00 + 14 hours = 22 00 on Monday **[M1]**

Add 8 days

$22\,00 on Tuesday (next week) [M1]$

Add 16 hours

$22\,00 + 16 hours = 38\,00$

$14\,00 on Wednesday [M1]$

Add 20 minutes

$Wednesday, 14\,20 [A1]$

[Back to Question] *[Total Marks: 7]*

17 **(a)** $920 \div 24$ **[M1]**

38.333... hours **[A1]**

38 hours 20 minutes **[A1]**

(b) Adds their time from part (a) to Wednesday 17 50 **[M1]**

Wednesday 17 50 + 20 minutes = 18 10 **[M1]**

Wednesday 18 10 + 38 hours = Thursday 18 10 + 14 hours **[M1]**

Day: Friday **[A1]**

Time: 08 10 **[A1]**

[Back to Question] *[Total Marks: 7]*

18 **(a)** $\frac{6600}{800}$ or 8.25 **[M1]**

8 hours 15 minutes **[A1]**

(b) Arrival time in City A time = 05 55 **[M1]**

00 55 Wednesday **[A1]**

(c) Departure from City B = 15 15 (local time) or 20 15 (City A time) **[M1]**

Arrival in City B time = 23 00 **[M1]**

04 00 Thursday **[A1]**

[Back to Question] *[Total Marks: 7]*

19 **(a)** $\dfrac{35}{x} + \dfrac{42}{x-5}$ **[B1]**

(b) 4 hours 33 minutes $= 4 + \dfrac{33}{60}$ or $\dfrac{91}{20}$ hours **[B1]**

$\dfrac{35}{x} + \dfrac{42}{x-5} = \dfrac{91}{20}$ **[M1]**

Multiplies by common denominator: $20 \times 35(x-5) + 20 \times 42x = 91x(x-5)$ **[M1]**

$700x - 3500 + 840x = 91x^2 - 455x$ **[M1]**

$1540x - 3500 = 91x^2 - 455x \implies 91x^2 - 1995x + 3500 = 0$ **[M1]**

Divides by 7 to reach $13x^2 - 285x + 500 = 0$ without errors **[A1]**

(c) Uses quadratic formula or factors: $(13x - 25)(x - 20) = 0$ **[M1]**

$x = 20$ or $x = \dfrac{25}{13}$ (or 1.92...) **[A1]**

$x = 20$ only (rejects $\dfrac{25}{13}$ as speed $x - 5$ must be positive) **[A1]**

[Back to Question] *[Total Marks: 8]*

20 **(a)** Minutes in a year $= 365 \times 24 \times 60$ **[M1]**

525600 minutes **[M1]**

Number of orbits $= \dfrac{525600}{92} = 5713.043\ldots$ **[M1]**

5713 **[A1]**

(b) **(i)** $\dfrac{11440}{880} = 13$ **[B1]**

(ii) Time difference $= 9 - (-4) = 13$ hours (Q is 13 hours ahead) **[M1]**

Departure time in City Q's local time $= 22:45$ (Monday) $+13$ hours $= 11:45$ (Tuesday) **[M1]**

Arrival time in City Q's local time $= 11:45$ (Tuesday) $+13$ hours (flight time) $= 00:45$ (Wednesday) **[A1]**

[Back to Question] *[Total Marks: 8]*

21 **(a)** 15×92.75 or 1380 min 675 s **[M1]**

23 h 11 min 15 s **[A1]**

(b) Arrival time in City A local time $= 09\,45 + 8$ h 20 min or $18\,05$ **[M1]**

$26\,05 - 18\,05$ **[M1]**

8 **[A1]**

(c) Distance travelled by Train P at 08:30 $= 90 \times 0.5 = 45$ km **[M1]**

Relative speed of approaching trains $= 90 + 120 = 210$ km/h **[M1]**

Time taken to meet from 08:30 $= (360 - 45) \div 210 = 1.5$ hours **[M1]**

10 00 **[A1]**

[Back to Question] *[Total Marks: 9]*